FINAL PROJECT REPORT

Automated Network Request Management in ServiceNow

Project Name	Automated Network Request Management in ServiceNow
Prepared By	
Platform	ServiceNow
Date	14 October 2026

Project Report Structure

• 1. Introduction
• 2. Ideation Phase
• 3. Requirement Analysis
• 4. Project Design
• 5. Project Planning & Scheduling
• 6. Functional and Performance Testing
• 7. Results
• 8. Advantages & Disadvantages
• 9. Conclusion
• 10. Future Scope
• 11. Appendix

1. INTRODUCTION

1.1 Project Overview

The Automated Network Request Management in ServiceNow is a workflow-automation prototype designed to streamline the lifecycle of network-related service requests. Employees or authorized requesters submit a structured Network Request through the ServiceNow Service Catalog. The request captures requester, connection, contact, address and reference information, creates the service request and requested item, and passes the request into Flow Designer for operational processing and approval.

The solution uses ServiceNow low-code capabilities rather than an external application stack. Catalog variables and a Requester Information variable set support structured data capture, while a Catalog UI Policy provides configured dynamic field behaviour. Flow Designer retrieves catalog variables, creates a configured Database Tables (u_database_tables) operational record, requests approval, evaluates the approval result, updates the record and invokes email notification actions. Roles and ACLs protect the operational data, and Flow Designer Execution Details/platform logs support troubleshooting and validation.

1.2 Purpose

• Reduce manual effort and human error in network-service request handling.
• Provide a structured, user-friendly Service Catalog entry point for network requests.
• Standardize request data capture and approval processing.
• Create a centralized operational record linked to the service-request lifecycle.
• Improve visibility through REQ/RITM tracking and automated notifications.
• Provide controlled access using roles and ACLs.
• Create a maintainable ServiceNow foundation that can be extended with fulfilment integrations, analytics and additional request categories.

2. IDEATION PHASE

2.1 Problem Statement

Employees and teams requesting network services need a faster, clearer and more traceable way to submit and follow requests. Manual or fragmented request handling can lead to incomplete information, repeated clarification, inconsistent approval routing, delayed handoffs and limited visibility into the current state of a request.

The operational problem is the movement of request information between submission, approval and fulfilment activities. IT administrators and fulfilment teams need standardized data, controlled access and a repeatable workflow that reduces manual intervention while preserving human approval decisions.

2.2 Empathy Map Canvas

Stakeholder	Think & Feel	See	Say & Do	Pain	Gain
End User / Requester	Wants a simple form, quick processing and clear status.	Service Portal, request forms, notifications.	Enters service details, submits request and tracks REQ/RITM.	Repeated data entry, unclear status, waiting for responses.	Structured submission, visibility and faster communication.
IT Administrator	Wants maintainable configuration and reliable automation.	Catalog, Flow Designer, tables, ACLs and execution details.	Configures variables, flows, security and troubleshooting.	Manual configuration effort and difficult diagnosis when flows fail.	Centralized configuration and execution visibility.
Network Fulfilment Team	Wants complete request data and clear handoffs.	Operational records, assignments and request details.	Reviews approved requests and performs fulfilment activity.	Incomplete data and avoidable clarification.	Standardized records and cleaner handoff.
Approver	Wants enough information to make a timely decision.	Approval record and request context.	Reviews and approves/rejects requests.	Unstructured requests and unclear approval context.	Controlled routing and audit-friendly approval handling.

2.3 Brainstorming

• Guided Network Request catalog item with standardized fields.

• Dynamic forms that show only relevant request information.

• Automatic retrieval of catalog variables into the workflow.

• Operational record creation from submitted request data.

• Approval routing and decision-based workflow branching.

• Automated request-created and approved/resolved notifications.

• REQ/RITM lifecycle visibility and audit-friendly tracking.

• Role/ACL-based protection of operational data.

• Future network-device/API fulfilment and advanced analytics.

The selected concept was the centralized ServiceNow request lifecycle because it addresses the highest-value operational handoffs while remaining feasible within the platform's low-code configuration model.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Requester Experience	System Response
Need Service	Employee identifies a network-service need.	Service Catalog provides a centralized entry point.
Open Portal	Requester opens ServiceNow Service Portal.	Networks and Connectivity category exposes the Network Request item.
Enter Details	Requester enters requested-for, mobile, connection, address and reference information.	Catalog variables and UI Policy control data capture and field behaviour.
Submit Request	Requester submits the completed form.	ServiceNow creates REQ/RITM and triggers Flow Designer.
Create Operational Record	Requester waits for processing.	Get Catalog Variables retrieves configured inputs and Create Database Tables Record maps them into u_database_tables.
Approval	Requester waits for approval decision.	Ask for Approval requests approval and the flow evaluates Approved versus alternate/rejected outcome.
Update & Notify	Requester receives progress/decision communication.	Operational record is updated and configured email actions send request-created or approved/resolved communication.
Track / Fulfil	Authorized users follow the request and perform fulfilment.	REQ/RITM and operational records retain lifecycle information.
Close	Request reaches its configured outcome.	Records and execution history provide traceability.

3.2 Solution Requirements

Requirement	Description
FR-1	Provide a Service Catalog Network Request item under Networks and Connectivity.
FR-2	Capture the configured requester, contact, connection, address and reference variables.
FR-3	Provide dynamic field behaviour using the configured Catalog UI Policy.
FR-4	Create a request/RITM and initiate Flow Designer processing.
FR-5	Retrieve catalog variables within Flow Designer.
FR-6	Create and populate the configured Database Tables operational record.
FR-7	Initiate approval and evaluate the approval outcome.
FR-8	Update Approval Status and other configured fields for approved/rejected outcomes.
FR-9	Send automated request-created and approved/resolved notifications.
FR-10	Support REQ/RITM lifecycle and request-status tracking.
FR-11	Protect operational records with configured roles and Create/Write/Read/Delete ACLs.
FR-12	Support troubleshooting through Flow Designer Execution Details and platform logs.

Non-functional requirements include usability, security, reliability, performance, availability, scalability, maintainability, data integrity, compatibility and auditability/troubleshooting.

3.3 Data Flow Diagram

Level 0 context: Requester → ServiceNow Network Request Management → Approver / Network Fulfilment / IT Administration, with request status and notifications returning through the platform.

Level 1 detailed flow: Capture Request → Retrieve & Validate Variables → Create Operational Record → Request Approval → Evaluate Decision → Update & Notify → Fulfilment / Closure. Primary stores are D1 Request/RITM and D2 Database Tables (u_database_tables).

3.4 Technology Stack

Component	Technology	Purpose
User Interface	ServiceNow Service Portal / Service Catalog	Requester-facing network request experience.
Catalog Configuration	Catalog Variables / Requester Information variable set / UI Policy	Structured data capture and dynamic form behaviour.
Workflow Automation	ServiceNow Flow Designer	Variable retrieval, record creation, approval, branching and notifications.
Request Records	ServiceNow Request / Requested Item	Lifecycle and service-request context.
Operational Data	Database Tables (u_database_tables)	Structured network-request operational record.
Approval	Ask for Approval / Approval Request relationship	Approval handling and relationship visibility.
Notifications	Flow Designer Send Email actions	Request-created and approved/resolved communication.
Security	Roles + ACLs	Controlled access to operational records.
Diagnostics	Flow Execution Details / platform logs	Workflow troubleshooting and validation.
Infrastructure	ServiceNow cloud / PDI	Execution and storage environment.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution fits the identified problem by moving network-service requests from fragmented/manual handling into a centralized, structured ServiceNow workflow. Requesters receive a guided form and lifecycle visibility; approvers receive a controlled approval action; administrators gain configuration and execution visibility; and fulfilment teams receive standardized operational data.

• Customer segment: employees/requesters, approvers, network fulfilment teams and IT administrators.

• Jobs-to-be-done: submit complete requests, obtain timely approval, process fulfilment and track outcomes.

• Constraints: current implementation is a ServiceNow prototype; external network-device/API integration is not implemented.

• Root causes: inconsistent request data, manual handoffs, repeated clarification and limited centralized visibility.

• Solution fit: Service Catalog + Flow Designer + operational record + approval + notification + security.

4.2 Proposed Solution

The proposed solution is an end-to-end ServiceNow-based network request workflow. It begins with a Network Request catalog item, captures structured variables, creates an operational record, requests approval, branches according to the decision, updates the record, and communicates status through automated notifications.

• Novelty: applies ServiceNow low-code workflow orchestration to a structured network-request lifecycle rather than treating the platform only as a ticket repository.

• Customer impact: simpler submission, standardized information, better status visibility and fewer manual handoffs.

• Operational impact: consistent mappings, approval traceability, controlled access and easier troubleshooting.

• Scalability: additional network services, approval chains, fulfilment tasks, integrations, dashboards and AI-assisted capabilities can be added incrementally.

4.3 Solution Architecture

Architecture layers: User Layer (Requester) → ServiceNow Application Layer (Service Portal, Service Catalog, Variables/UI Policy, Flow Designer) → Data & Control Layer (REQ/RITM, Database Tables, Approval Request, Roles/ACLs) → Operations Layer (Fulfilment, Notifications, Execution Details). Future extensions include network-device APIs, advanced analytics and external messaging.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Epic / Focus	Story Points	Release Outcome
Sprint 1	Catalog & Form Setup	9	Catalog item, variables, Requester Information variable set and UI Policy.
Sprint 2	Workflow, Approval & Data	15	Flow Designer, variable retrieval, operational record mapping and approval.
Sprint 3	Notifications, Security & Fulfilment	13	Email actions, ACLs, approval relationship, lifecycle and fulfilment handoff.
Sprint 4	Testing, Debugging & Deployment	16	Functional/UAT testing, execution diagnostics, security validation and documentation.

Total planned/completed story points: 53. Average velocity across four sprints: 13.25 story points per sprint.

The planning model is adapted from the supplied final-report structure, while the sprint content reflects the Network Request implementation rather than the Metro ticketing workflow.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test ID	Test Scenario	Final Result
FT-01	Network Request catalog/form validation	Pass
FT-02	Dynamic field behaviour / UI Policy	Pass
FT-03	Request submission and REQ/RITM creation	Pass
FT-04	Catalog variable retrieval	Pass
FT-05	Variable-to-operational-record mapping	Pass
FT-06	Approval processing and decision branching	Pass
FT-07	Request-created notification	Pass
FT-08	Approved/resolved or rejection notification path	Pass
FT-09	Request status/lifecycle tracking	Pass
FT-10	Role and ACL access	Pass
FT-11	Approval Request relationship	Pass
FT-12	End-to-end workflow execution	Pass
PT-01	Portal response behaviour	Pass - no abnormal delay reported in project testing
PT-02	Repeated request execution consistency	Pass
PT-03	Workflow execution consistency	Pass
PT-04	Portal/form compatibility	Pass

Machine-learning, GenAI, Power BI and Tableau model-performance metrics are not applicable to the current implementation because this solution does not use a trained AI/ML model or BI platform. These were documented as N/A rather than inventing model or dashboard metrics.

7. RESULTS

7.1 Output

The final implementation demonstrates a working ServiceNow Network Request workflow. The delivered result includes the Network Request catalog item, structured request variables, dynamic UI behaviour, REQ/RITM creation, Flow Designer automation, operational record creation, approval handling, notifications, request tracking, role/ACL protection and troubleshooting support.

Output Area	Implemented Result
Service Catalog	Network Request under Networks and Connectivity with configured variables.
Request Lifecycle	REQ/RITM creation and visible lifecycle stages.
Automation	Flow Designer retrieves variables, creates record, requests approval and branches on outcome.
Operational Data	Database Tables (u_database_tables) record populated from catalog values.
Approval	Ask for Approval with Approved / alternate decision handling.
Notifications	Request Created and Request Resolved email actions configured.
Security	Four active custom-table ACLs for Create, Write, Read and Delete.
Traceability	Approval Request relationship and Flow Execution Details support validation.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages / Current Limitations
Reduces repetitive manual request handling.	Current implementation is a project prototype rather than a production network-operations deployment.
Standardizes network-request data capture.	Automated network-device/API fulfilment is not implemented.
Uses ServiceNow low-code automation for maintainability.	Advanced SLA/escalation and analytics are future scope.
Provides centralized REQ/RITM and operational-record visibility.	External SMS/WhatsApp and other messaging channels are future scope.
Supports controlled approval and ACL-based access.	No production-scale load benchmark figures were supplied.
Provides execution details for troubleshooting.	Current flow still depends on configured human approval decisions.

9. CONCLUSION

The Automated Network Request Management project demonstrates how ServiceNow can be used to implement a structured, low-code workflow for network-service requests. The project integrates Service Catalog interaction, dynamic form behaviour, Flow Designer automation, operational data storage, approval handling, notifications, request tracking and access control into one coherent lifecycle.

The main implementation focus was the reliability of the request handoffs: capturing complete data, mapping it correctly into the operational record, routing approval, evaluating the decision and preserving traceability. The resulting prototype provides a practical foundation for future network fulfilment automation and enterprise integration.

10. FUTURE SCOPE

- Integration with network-device and infrastructure-management APIs for automated fulfilment.
- Automated task creation, assignment and closure for network operations.

• Multi-level manager/security approval chains and policy-driven routing.

• SLA tracking, escalation and advanced operational dashboards.

• External email/SMS/WhatsApp messaging channels.

• AI-assisted request classification, priority recommendation and anomaly detection.

• Advanced network analytics and capacity/demand reporting.

• Additional network-service catalog items and reusable request templates.

• Production deployment hardening, monitoring and performance benchmarking.

11. APPENDIX

Appendix A - Functional Data Mapping

Data Element	Source	Destination	Purpose
Requested For	Catalog variable	Database Tables → Requested For	Identifies the employee for whom the request is submitted.
Mobile Number	Catalog variable	Database Tables → Mobile Number	Stores contact information.
Type of Connection	Catalog variable	Database Tables → Type of Connection	Captures New/Existing connection type.
Existing ID	Catalog variable / conditional field	Configured request context	Stores existing connection reference when applicable.
Total Amount	Catalog variable	Database Tables → Total Amount	Stores configured request amount.
Mode of Payment	Catalog variable	Database Tables → Mode of Payment	Stores selected payment-mode value.
Address	Catalog variable	Database Tables → Customer Address	Stores requested service/customer address.
Database Number	Catalog variable	Database Tables → Database Number	Stores configured database/reference number.
Request Number	Catalog variable / request context	Database Tables → Request Number	Supports request identification and tracking.
Approval Status	Flow / approval result	Database Tables → Approval Status	Tracks Requested, Approved or Rejected state.
REQ/RITM	ServiceNow request records	Request / Requested Item	Preserves service-request context and lifecycle tracking.

Appendix B - Current Implementation Boundary

Implemented: Service Catalog, catalog variables, Requester Information variable set, Catalog UI Policy, Flow Designer, Database Tables operational record, REQ/RITM tracking, Ask for Approval, email actions, ACLs, Approval Request relationship and execution diagnostics.

Future / Not Implemented: automated network-device APIs, external messaging channels, advanced analytics/BI, AI/ML model, automated payment processing and production-scale performance benchmarking.